

Development of a Deep Reinforcement Learning Agent for Optimal Project Portfolio Budgeting under Uncertain Cash Flows

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Abstract

Project portfolio budget allocation for EPC contractors in the oil and gas sector is a sequential decision-making problem under uncertain cash inflows, cash outflows, and project performance. Each portfolio consists of multiple projects with heterogeneous attributes, including budget at completion, start and finish dates, contractual S-curves, fixed profit margins, milestone-based payment structures, and uncertain contractor performance represented through Earned Value Management metrics such as cumulative SPI and CPI. The model also captures delayed and uncertain milestone payments, working-capital shortages, credit usage during negative-cashflow periods, and management intervention actions that affect cashflow and performance trajectories.

Despite its practical importance, rigorous mathematical modeling of project portfolio budgeting under joint cashflow and performance uncertainty remains largely absent in recent literature, while most AI/ML-related studies discuss the topic qualitatively. This paper proposes a custom stochastic reinforcement learning environment for learning adaptive budget allocation policies under unknown future uncertainty. A PPO-based agent is trained with a multi-objective reward function to allocate constrained budgets efficiently, prioritize economically viable projects, avoid loss-making allocations, and learn optimal allocation timing. The framework provides a foundational RL model for EPC oil and gas portfolio budgeting, supports future fine-tuning with anonymized industry data, and enables immediate next-period allocation recommendations as well as rolling 12-period trajectory simulation.

Keywords: Project Portfolio Budgeting, Reinforcement Learning, EPC Contractors, Oil and Gas Projects, Cashflow Uncertainty, Earned Value Management, PPO, Rolling Horizon Control

1 Introduction

1.1 Motivation and Context

Project portfolio management (PPM) is a fundamental organizational capability for maintaining competitive advantage and achieving strategic objectives. In project-based industries, portfolio decisions determine which initiatives are selected, prioritized, and funded

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under limited resources, conflicting criteria, and uncertain execution outcomes (Archer and Ghasemzadeh, 1999; Cooper et al., 2001; Martinsuo, 2013; Killen et al., 2008). These decisions become particularly critical in large-scale engineering environments, where organizations must coordinate multiple concurrent projects while balancing strategic value, financial feasibility, schedule pressure, and risk exposure.

In the Engineering, Procurement, and Construction (EPC) sector, especially in oil and gas projects, portfolio budgeting is complicated by heterogeneous contract structures, long project durations, capital intensity, and uncertain cash-flow behavior (Merrow, 2011; Cui et al., 2010; Kenley and Wilson, 1986; Kaka and Price, 1993). Budget allocation must account for both cash outflows associated with procurement, labor, equipment, and sub-contractors, and delayed cash inflows tied to contractual milestones and client approvals. Consequently, portfolio budgeting is not merely a one-time allocation problem; it is a sequential decision process in which current funding choices influence future project progress, liquidity conditions, and portfolio-level performance.

This challenge is amplified by persistent uncertainty in project execution. Large industrial and infrastructure projects frequently experience delays and cost overruns, and such deviations can propagate through both schedules and financial plans (Flyvbjerg et al., 2002, 2003; Love et al., 2013, 2016). Earned value indicators such as the Schedule Performance Index (SPI) and Cost Performance Index (CPI) provide a compact representation of execution performance and can support project monitoring and control (Fleming and Koppelman, 2016; Lipke et al., 2009; Vanhoucke, 2012). However, these indicators also underscore the difficulty of making reliable budget allocation decisions when future project states, payment realizations, and performance deviations remain uncertain. Therefore, portfolio budgeting under uncertain cash flows requires methods that can reason across multiple projects, multiple time periods, and stochastic state transitions.

The broader literature on project portfolio management similarly characterizes such environments by large data volumes, multiple decision criteria, resource constraints, and the need for investment optimization. These characteristics are consistent with prior research on project portfolio selection, project scheduling under uncertainty, and construction cash-flow forecasting (Gutjahr and Froeschl, 2013; Doerner et al., 2006; Khanzadi et al., 2018; Golenko-Ginzburg and Gonik, 1997; Ballestín and Blanco, 2015; Barraza et al., 2007). Taken together, they motivate the development of intelligent decision-support methods when static or manually driven budgeting approaches become insufficient.

1.2 Research Gap

A substantial body of literature has addressed project portfolio selection, prioritization, and scheduling through operations research, multi-criteria decision-making, stochastic programming, and metaheuristic optimization (Archer and Ghasemzadeh, 1999; Liesiö et al., 2007; Klerides and Hadjiconstantinou, 2019; Elshaer, 2014). These methods provide valuable structure for complex decision problems, but they are often less suitable when portfolio budget allocation must respond to evolving project conditions under uncertainty. In many existing formulations, project cash flows, budget requirements, and performance trajectories are treated as fixed inputs, deterministic plans, or exogenous scenarios. This limits their ability to capture the interaction between current allocation decisions and future project and financial states.

A central limitation in this context is that portfolio budgeting in EPC environments is inherently sequential. Budget releases made at one time step affect project execution, milestone realization, cash availability, and subsequent allocation opportunities. Yet much of the existing literature approaches project portfolio decisions as static optimization,

stage-wise planning, or scheduling problems rather than as a learned sequential policy problem under uncertainty. As a result, the adaptive linkage between observed portfolio state and subsequent allocation action remains insufficiently modeled.

Recent advances in machine learning and reinforcement learning provide a promising direction for this class of problems. Reinforcement learning is specifically designed for sequential decision-making under uncertainty and has demonstrated value in operations, scheduling, and process control settings (Puterman, 1994; Sutton and Barto, 2018; Wauters et al., 2016; Chen et al., 2018; Hubbs et al., 2020; Bengio et al., 2021). Nevertheless, its application to project portfolio budgeting under uncertain cash flows remains limited. There is still a need for a reproducible framework that can integrate project-level uncertainty, cash-flow timing, milestone-based payments, and adaptive resource allocation within a portfolio-level sequential decision architecture.

A further practical constraint is data availability. Detailed industrial project portfolio data are typically confidential and difficult to release publicly, which creates a barrier to reproducible experimentation. As a result, the literature lacks sufficiently transparent benchmark environments for testing adaptive portfolio budgeting policies. This gap motivates the use of literature-calibrated synthetic data as a practical foundation for developing and evaluating learning-based decision-support systems (Jordon et al., 2022; Xu et al., 2019).

1.3 Proposed Approach and Contributions

To address these gaps, this paper develops a deep reinforcement learning framework for optimal project portfolio budgeting under uncertain cash flows. The problem is modeled as a sequential decision process in which an agent observes the evolving state of a project portfolio and allocates constrained financial resources over time. The state representation incorporates project-level and portfolio-level information, including budget at completion, planned duration, contractual cash-flow patterns, performance indicators, milestone-based payment structures, and liquidity-related variables.

The framework is designed to learn an allocation policy directly from repeated interaction with a stochastic simulation environment. Rather than relying on predefined deterministic decision rules or repeated re-optimization procedures, the agent learns how present allocation choices affect future portfolio states and cumulative performance over time. To learn this policy, the study employs deep reinforcement learning, with Proximal Policy Optimization (PPO) as the primary training algorithm (Schulman et al., 2017), consistent with the broader reinforcement-learning literature on stochastic control and policy optimization (Sutton and Barto, 2018; Haarnoja et al., 2018).

After training, the learned policy can also be deployed within the simulation environment to generate a forward policy rollout over a specified horizon, such as the next 12 time steps, in order to illustrate plausible portfolio evolution and corresponding recommended allocations. This forward policy rollout should be interpreted as a simulated policy-driven trajectory rather than as a realized future observation. Its purpose is to demonstrate the behavioral implications of the learned allocation policy under the modeled environment dynamics until actual future project states become available.

Because real EPC portfolio datasets are typically inaccessible due to confidentiality restrictions, the framework uses a literature-calibrated synthetic data generation process. This allows reproducible experimentation while preserving consistency with known characteristics of project cash flows, schedule variability, and portfolio structure. The synthetic environment is intended as a research testbed that can later be refined and calibrated using anonymized organizational data.

The main contributions of this paper are as follows:

1. **A sequential formulation of project portfolio budgeting:** The study formulates EPC portfolio budgeting under uncertain cash flows as a dynamic decision problem in which allocations, project progress, and liquidity conditions evolve jointly over time.
2. **A custom reinforcement learning environment for portfolio budgeting:** The paper develops an environment that captures project-level uncertainty, cash inflows and outflows, milestone-based payments, and portfolio-level financial constraints.
3. **A deep reinforcement learning policy for adaptive allocation:** The framework trains an agent to learn budget allocation decisions directly from stochastic interaction with the environment.
4. **A forward policy rollout capability for trajectory demonstration:** The study shows how the trained policy can be used to simulate plausible future portfolio trajectories and recommended allocations over a finite horizon.
5. **A literature-calibrated synthetic data framework:** The study provides a reproducible synthetic portfolio-generation procedure to address the industrial data confidentiality barrier.

1.4 Paper Organization

The remainder of this paper is organized as follows. Section 2 reviews prior research on project portfolio management, project control under uncertainty, and reinforcement learning for sequential decision-making. Section 3 formulates the project portfolio budgeting problem under uncertain cash flows as a sequential stochastic decision problem. Section 4 presents the proposed solution methodology, including the synthetic data generation framework, cash-flow and uncertainty modeling, reinforcement learning environment design, and PPO-based training procedure. Section 5 reports the computational results and analysis, including benchmark comparisons, robustness assessment, and forward policy rollout analysis. Finally, Section 6 concludes the paper and outlines future research directions.

2 Literature Review

2.1 Project Portfolio Management Under Uncertainty

Project portfolio management (PPM) has been extensively studied in operations research and management science. Early work by Archer and Ghasemzadeh (1999) established frameworks for project selection and prioritization, while Cooper et al. (2001) developed stage-gate processes for new product development portfolios. However, these approaches primarily address the *selection* problem (which projects to pursue) rather than the *execution* problem (how to allocate resources dynamically).

? formulated multi-objective portfolio optimization models balancing expected return, risk, and strategic alignment. Doerner et al. (2006) applied Pareto ant colony optimization to generate efficient frontiers for portfolio selection. These methods assume deterministic project execution once selected, ignoring the sequential decision-making required under performance uncertainty.

Stochastic programming approaches have been proposed to handle uncertainty. [Golenko-Ginzburg and Gonik \(1997\)](#) modeled resource-constrained project scheduling with stochastic activity durations. [Klerides and Hadjiconstantinou \(2019\)](#) developed two-stage stochastic programs for portfolio selection under demand uncertainty. However, these methods require explicit scenario trees and known probability distributions, which are unavailable for contractor performance dynamics.

2.2 Project Performance Uncertainty and Empirical Evidence

A substantial body of empirical research documents systematic cost and schedule overruns in capital projects. [Flyvbjerg et al. \(2002\)](#) analyzed 258 transportation infrastructure projects, finding mean cost overruns of 28% and schedule delays of 27%. [Flyvbjerg et al. \(2003\)](#) extended this analysis to 258 projects across multiple sectors, confirming left-skewed performance distributions with long tails toward poor outcomes.

[Merrow \(2011\)](#) provided the most comprehensive analysis of EPC project performance, examining 318 industrial megaprojects in oil & gas, chemicals, and mining. Key findings include: (1) mean SPI at completion of 0.81 (19% behind schedule), (2) mean CPI of 0.87 (15% over budget), (3) positive correlation between schedule and cost performance ($\rho \approx 0.68$), and (4) systematic variation in performance by project size, complexity, and contractor experience.

[Love et al. \(2013\)](#) fitted probability distributions to schedule overrun data from 276 construction projects, validating Beta and lognormal distributions as appropriate models. [Love et al. \(2016\)](#) analyzed the relationship between SPI and CPI, confirming that schedule delays typically precede cost overruns due to extended overhead and acceleration costs.

2.3 Cashflow Modeling and S-Curves

Project cashflow modeling has a rich history in construction management. [Kenley and Wilson \(1986\)](#) introduced the idiographic approach to cashflow forecasting, recognizing that each project exhibits unique spending patterns. [Kaka and Price \(1993\)](#) developed neural network models for cashflow prediction based on historical project data.

The S-curve model, representing cumulative spending as a sigmoid function of project duration, has become the standard representation. [Cioffi \(2005\)](#) provided analytical parameterization of S-curves using Beta distributions, enabling closed-form expressions for cashflow velocity and peak spending timing. [Barraza et al. \(2007\)](#) extended this to probabilistic S-curves incorporating uncertainty in project duration and spending patterns.

[Cui et al. \(2010\)](#) analyzed milestone-based payment structures in construction contracts, finding that 87% of contracts use progress-based payments rather than time-based schedules. This finding motivates our modeling choice to link payment timing to actual progress (SPI) rather than calendar dates.

2.4 Reinforcement Learning in Operations Research

RL has emerged as a powerful tool for sequential decision-making under uncertainty. [Sutton and Barto \(2018\)](#) provide the foundational textbook covering Markov Decision Processes (MDPs), temporal difference learning, and policy gradient methods. Recent advances in deep RL, particularly Proximal Policy Optimization (PPO) ([Schulman et al., 2017](#)) and Soft Actor-Critic (SAC) ([Haarnoja et al., 2018](#)), have enabled application to high-dimensional continuous control problems.

Applications of RL to operations research problems have grown rapidly. [Hubbs et al. \(2020\)](#) applied deep RL to chemical production scheduling, achieving near-optimal solutions with 1000x speedup over MIP. [Bengio et al. \(2021\)](#) surveyed machine learning for combinatorial optimization, highlighting RL’s ability to learn heuristics that generalize across problem instances.

In project management, RL applications remain limited. [Wauters et al. \(2016\)](#) used RL for resource-constrained project scheduling with deterministic durations. [Chen et al. \(2018\)](#) applied RL to construction project scheduling under resource uncertainty. However, no prior work addresses portfolio-level budgeting under cashflow uncertainty with stochastic contractor performance.

2.5 Synthetic Data Generation for OR/ML Research

The use of synthetic data in operations research has a long tradition. [Kolisch and Sprecher \(1997\)](#) created the Project Scheduling Problem Library (PSPLIB), a benchmark dataset of synthetic instances that became the standard for algorithm comparison. [?](#) generated synthetic project networks to study schedule risk analysis methods.

In machine learning, synthetic data generation has become essential for domains with data scarcity or privacy constraints. [Jordon et al. \(2022\)](#) surveyed synthetic data generation methods, emphasizing the importance of preserving statistical properties of real data. [Xu et al. \(2019\)](#) developed generative models for tabular data that maintain correlations and distributional characteristics.

For project management, [Vanhoecke \(2012\)](#) generated synthetic project data with controlled complexity levels to benchmark scheduling algorithms. [Elshaer \(2014\)](#) created synthetic portfolios with varying risk profiles to test portfolio optimization methods. Our work extends this tradition by developing a comprehensive synthetic data generator calibrated to empirical distributions from multiple literature sources, ensuring realism while maintaining reproducibility.

2.6 Research Gaps

Despite extensive research in project portfolio management, performance uncertainty, and RL applications, critical gaps remain:

1. **No RL framework for portfolio budgeting:** Existing RL applications to project management focus on single-project scheduling, not portfolio-level resource allocation under cashflow uncertainty.
2. **Lack of realistic synthetic environments:** Prior synthetic data generators for project management do not model stochastic contractor performance, milestone-based payments, or portfolio-level cashflow dynamics.
3. **Absence of rolling horizon RL architectures:** Rolling horizon control has not been integrated with RL for portfolio management, despite its natural fit for handling variable-duration portfolios.
4. **No methodology for literature-calibrated data generation:** Existing approaches either use proprietary data (unavailable for research) or arbitrary synthetic distributions (lacking empirical grounding).

This paper addresses these gaps by developing the first RL framework for portfolio budgeting with a literature-calibrated synthetic environment and rolling horizon architecture.

3 Problem Formulation

This formulation provides a novel representation of project portfolio budget allocation that integrates mandatory minimum funding requirements, soft budget constraints with endogenous credit dynamics, performance-triggered corrective interventions, and milestone-based revenue recognition within a unified stochastic, multi-period optimization framework. Unlike traditional project portfolio selection models [Archer and Ghasemzadeh \(1999\)](#); [Liesiö et al. \(2007\)](#); [Gutjahr and Froeschl \(2013\)](#), the focus is on operational control of committed projects rather than ex-ante selection. The formulation extends earned-value management frameworks [Anbari \(2003\)](#); [Fleming and Koppelman \(2016\)](#); [Lipke et al. \(2008\)](#) by embedding them within a sequential decision framework with explicit financial accounting, soft budget constraints, endogenous intervention mechanisms, and multi-objective performance evaluation, while remaining solution-agnostic and compatible with diverse solution methodologies including reinforcement learning, approximate dynamic programming, or scenario-based stochastic optimization.

3.1 Management Problem and Portfolio Definition

We consider a stochastic, multi-period budget allocation problem for a fixed portfolio of contracted projects [Martinsuo \(2013\)](#). The portfolio is defined as the aggregation of a finite set of projects,

$$\mathcal{P} = \{1, 2, \dots, n\},$$

where each project $i \in \mathcal{P}$ represents a contractual undertaking with its own budget, planned schedule, expenditure profile, payment structure, and performance state. The portfolio is not a decision variable in the sense of project selection [Archer and Ghasemzadeh \(1999\)](#); [Cooper et al. \(2001\)](#). Instead, it is the set of ongoing and committed projects whose execution must be managed jointly under limited financial and operational resources.

The planning horizon is indexed by discrete periods $t \in \mathcal{T} = \{1, \dots, H\}$. At each period, the decision-maker observes the current portfolio condition and allocates financial resources across active projects [Lambrechts et al. \(2008\)](#); [Vanhoucke \(2012\)](#). Projects consume resources over time, generate cash inflows through milestone-based payments, and may deviate from their contractual plans because of schedule delays, cost overruns, funding shortages, payment delays, contractor performance uncertainty, or operational disruptions [Flyvbjerg et al. \(2003\)](#); [Merrow \(2011\)](#).

The central management problem is to determine a sequential allocation policy that balances multiple objectives: completing projects close to their planned finish dates, preserving profitability, avoiding excessive borrowing, reducing delay penalties, limiting wasteful expenditure on economically unattractive projects, and maintaining acceptable portfolio-level financial health [Liesiö et al. \(2007\)](#). Different organizations may prioritize these goals differently.

A key feature of the problem is that budget allocation decisions do not produce known deterministic performance outcomes. After resources are allocated in period t , the contractor's realized response is uncertain and is only observed at the next reporting epoch through project-control information such as realized progress, earned value, schedule performance, cost performance, and updated remaining-duration estimates [Anbari \(2003\)](#); [Fleming and Koppelman \(2016\)](#). This makes the problem naturally sequential and feedback-driven under uncertainty.

3.2 Contractual Project Attributes

Each project $i \in \mathcal{P}$ is described by contractual and planned attributes known at the beginning of the planning process. Let s_i and f_i denote the planned start and planned finish periods, respectively. The planned duration is

$$D_i^{\text{plan}} = f_i - s_i + 1,$$

and the contractual planned execution window is $\mathcal{A}_i^{\text{plan}} = \{t : s_i \leq t \leq f_i\}$.

The budget at completion of project i is denoted by BAC_i . This value represents the planned cost basis of the project and is used to define planned expenditure, contract revenue, and credit exposure. The contract revenue is $CR_i = \text{BAC}_i(1 + \pi_i)$, where π_i is the contractual profit margin.

3.3 Planned Expenditure and Contractual S-Curve

The planned cumulative expenditure of each project follows a normalized S-curve [Barraza et al. \(2004\)](#). The normalized planned time is $\tau_{i,t} = (t - s_i + 1)/D_i^{\text{plan}}$ for $t \in \mathcal{A}_i^{\text{plan}}$, with truncation $\tau_{i,t}^c = \min\{1, \max\{0, \tau_{i,t}\}\}$.

The planned cumulative expenditure fraction is modeled as $F_i(\tau) = I_\tau(a_i, b_i)$, where $I_\tau(\cdot, \cdot)$ is the regularized incomplete Beta function and $a_i, b_i > 0$ are shape parameters. The planned cumulative expenditure is $\bar{E}_{i,t} = \text{BAC}_i F_i(\tau_{i,t}^c)$, and the planned period expenditure is $\bar{e}_{i,t} = \bar{E}_{i,t} - \bar{E}_{i,t-1}$. This S-curve represents the contractual baseline spending trajectory used for measuring planned progress, schedule deviation, and performance indices [Anbari \(2003\)](#).

3.4 Decision Variables and Constraints

The decision-maker's choice at each period t is the allocation of budget to each active project. Let $x_{i,t} \geq 0$ denote the budget allocated to project i at time t . A project is considered active if its cumulative progress $P_{i,t-1}^{\text{actual}} < 1$. The allocation decision is constrained by project-level mandatory minimum requirements and portfolio-level budget availability.

3.4.1 Mandatory Minimum Allocation Constraint

Each project i at time t has a minimum mandatory allocation $m_{i,t} \geq 0$ reflecting contractual obligations, debt-service requirements, or minimum execution commitments [Al-Sedairy \(2001\)](#); [Project Management Institute \(2017\)](#). The allocation must satisfy

$$x_{i,t} \geq m_{i,t}, \quad \forall i \in \mathcal{P}, t \in \mathcal{T}.$$

This constraint ensures that projects receive at least the minimum required funding to meet operational, contractual, or financial obligations in each period. If a project uses credit or triggers corrective intervention mechanisms, future mandatory allocations may increase to reflect additional cashflow requirements.

3.4.2 Soft Budget Constraint and Credit Mechanism

The portfolio has an available internal budget B_t^{avail} at each period t . If the total allocation across active projects exceeds the available internal budget, the excess is financed through

external credit. Let

$$c_t = \max \left(0, \sum_{i \in \mathcal{P}: P_{i,t-1}^{\text{actual}} < 1} x_{i,t} - B_t^{\text{avail}} \right)$$

denote the credit used at time t . Credit is allowed but creates interest-bearing debt that increases future cashflow obligations [Al-Jibouri \(2002\)](#); [Flyvbjerg \(2014\)](#).

Outstanding debt evolves according to

$$D_{t+1} = (1 + r_t)D_t + c_t - q_t,$$

where D_t is the debt at the start of period t , $r_t \geq 0$ is the interest rate, and $q_t \geq 0$ is the repayment made in period t . If repayment is not separately modeled, the debt accumulates as $D_{t+1} = (1 + r_t)D_t + c_t$. The credit mechanism is thus modeled as a soft budget constraint: allocations exceeding internal funds are feasible but impose financial penalties through interest charges.

3.5 Project State and Performance Dynamics

3.5.1 Project State Variables

At each period t , project i is described by a state vector including planned cumulative progress $P_{i,t}^{\text{plan}} \in [0, 1]$, actual cumulative progress $P_{i,t}^{\text{actual}} \in [0, 1]$, actual cumulative expenditure $E_{i,t} \geq 0$, schedule performance index $\text{SPI}_{i,t}$, cost performance index $\text{CPI}_{i,t}$, and remaining-duration estimate $RD_{i,t}$ [Anbari \(2003\)](#); [Lipke et al. \(2008\)](#). The performance gap is defined as

$$\Delta_{i,t} = P_{i,t}^{\text{plan}} - P_{i,t}^{\text{actual}}.$$

3.5.2 Performance-Triggered Corrective Intervention Mechanism

Following earned-value management practice [Fleming and Koppelman \(2016\)](#); [Project Management Institute \(2017\)](#), each project is continuously monitored by the owner's representative or project management consultant (PMC). When the performance gap exceeds a contractually defined threshold τ_i , a corrective intervention is automatically initiated [Al-Sedairy \(2001\)](#); [Ijaola and Irein \(2013\)](#). The intervention indicator is defined as

$$I_{i,t} = \begin{cases} 1, & \Delta_{i,t} > \tau_i, \\ 0, & \Delta_{i,t} \leq \tau_i. \end{cases}$$

The intervention is not a decision variable. It is an endogenous, state-dependent corrective mechanism representing contractual and governance-driven responses such as recovery planning, escalation, closer monitoring, or contractor corrective action [Hällgren and Maaninen-Olsson \(2009\)](#); [Love et al. \(2011\)](#). The intervention can be activated only if sufficient budget has been allocated to cover the associated management effort and potential acceleration costs. If $x_{i,t}$ falls below the threshold necessary to support intervention activities, the intervention may fail to activate even when $\Delta_{i,t} > \tau_i$.

3.5.3 Effect of Intervention on Future Performance

When intervention is triggered, it affects future project performance through two channels: increased near-term cashflow requirements and improved budget-to-progress conversion

efficiency Williams et al. (2009). Let $\alpha_{i,t} \geq 0$ denote the baseline efficiency with which allocated budget translates into progress. The intervention-adjusted efficiency is

$$\tilde{\alpha}_{i,t+1} = \alpha_{i,t+1}(1 + \kappa_{i,t}I_{i,t}),$$

where $\kappa_{i,t} \geq 0$ is the intervention effectiveness parameter. This parameter may be deterministic or stochastic, reflecting the uncertain success of recovery plans observed in practice Ijaola and Irefin (2013); Love et al. (2011).

The actual progress at $t + 1$ evolves as

$$P_{i,t+1}^{\text{actual}} = P_{i,t}^{\text{actual}} + \tilde{\alpha}_{i,t}x_{i,t} + \varepsilon_{i,t},$$

where $\varepsilon_{i,t}$ is a stochastic disturbance capturing contractor performance uncertainty.

The intervention also increases the mandatory minimum allocation for subsequent periods to reflect additional cashflow obligations:

$$m_{i,t+1} = m_{i,t+1}^0 + \chi_i I_{i,t},$$

where $m_{i,t+1}^0$ is the baseline mandatory allocation and $\chi_i \geq 0$ captures the additional required funding induced by the intervention.

The allocation decision $x_{i,t}$ influences intervention only indirectly: by affecting actual progress $P_{i,t}^{\text{actual}}$, it influences the performance gap $\Delta_{i,t}$, which in turn determines whether the intervention threshold is breached. The decision-maker can thus learn to allocate budget strategically to either avoid intervention costs or deliberately trigger interventions when the expected performance gain justifies the expenditure.

3.6 Revenue Recognition and Milestone-Based Payments

Revenue is recognized through contractual milestone payments tied to verified physical progress Barraza et al. (2004). Each project i has a set of milestones $\mathcal{M}_i = \{1, \dots, M_i\}$. Milestone $j \in \mathcal{M}_i$ is associated with a progress threshold $\theta_{i,j} \in (0, 1]$ and a payment fraction $\phi_{i,j} > 0$ such that $\sum_{j \in \mathcal{M}_i} \phi_{i,j} = 1$. When actual progress reaches $\theta_{i,j}$, the milestone payment

$$R_{i,j} = \phi_{i,j}CR_i(1 - \rho_i)$$

is received, where $\rho_i \in [0, 1)$ is the retention fraction. Retention is released upon project completion. Let V_t denote the total milestone-based revenue received in period t .

3.7 Objective Function

The objective is to maximize the expected cumulative discounted performance across the planning horizon. Let $\gamma \in (0, 1]$ denote the discount factor. The period performance metric aggregates milestone-based revenue, delay penalties, credit costs, and other portfolio-level costs:

$$R_t = V_t - \alpha_D D_t - C_t^{\text{credit}} - C_t^{\text{delay}} - C_t^{\text{other}},$$

where $\alpha_D \geq 0$ weights the portfolio delay (total periods all projects are behind schedule), C_t^{credit} represents interest and financing costs, C_t^{delay} captures contractual liquidated damages, and C_t^{other} includes other penalties or costs.

The total objective is

$$\max \mathbb{E} \left[\sum_{t=1}^H \gamma^{t-1} R_t \right],$$

where the expectation is taken over the stochastic evolution of project states, contractor responses, and intervention outcomes. The allocation policy must respect the mandatory minimum allocation constraints and the soft budget constraint with endogenous credit dynamics.

3.8 Notation and Parameters

Table 1 provides a comprehensive list of notation and parameters used in the problem formulation.

3.9 Scope and Modeling Assumptions

The formulation is subject to the following scope definitions and modeling assumptions:

- The portfolio composition is fixed. Project selection and termination decisions are not modeled.
- All projects are assumed to have been properly contracted with defined scopes, budgets, schedules, and payment structures before the planning horizon begins.
- The formulation is solution-agnostic. It can be embedded within a stochastic optimization framework, approximate dynamic programming, or a sequential decision-making environment for solution via reinforcement learning or other methods.
- Progress is measured as cumulative physical completion and is normalized to $[0, 1]$. Earned value is computed as the product of progress and budget at completion.
- Retention is withheld from interim progress payments and fully released with the final milestone at project completion.
- A separate defects liability period is not modeled because it lies beyond the active execution-focused planning horizon.
- Credit use is induced by allocation decisions exceeding available internal budget. Credit limits may be enforced through penalty terms in the objective function.
- Corrective interventions are automatically triggered by performance gaps exceeding contractually defined thresholds. The intervention's effectiveness is uncertain and may be represented through stochastic parameters or disturbance terms.
- Intervention-related costs are reflected through increased mandatory minimum allocations in subsequent periods and through the credit mechanism if additional budget must be borrowed.
- The formulation defines the decision structure, constraints, state dynamics, and performance evaluation. Numerical solution requires specification of the stochastic processes governing contractor response, intervention effectiveness, and project state evolution.

Table 1: Notation and Parameters

Name	Description	Mathematical Formulation
<i>Sets and Indices</i>		
Project set	Set of projects	$\mathcal{P} = \{1, \dots, n\}$
Time-period set	Set of time periods	$\mathcal{T} = \{1, \dots, H\}$
Project index	Index of a project	$i \in \mathcal{P}$
Time-period index	Index of a time period	$t \in \mathcal{T}$
<i>Decision Variable</i>		
Budget allocation	Budget allocated to project i at time t	$x_{i,t}$
<i>Contractual Attributes</i>		
Planned start and finish	Planned start and finish periods for project i	s_i, f_i
Planned duration	Planned duration of project i	$D_i^{\text{plan}} = f_i - s_i + 1$
Budget at completion	Total planned project budget	BAC_i
Profit margin	Contractual profit margin for project i	π_i
Contract revenue	Total contract revenue for project i	$CR_i = \text{BAC}_i(1 + \pi_i)$
Retention fraction	Retained fraction of project revenue	ρ_i
<i>Planned Expenditure (S-Curve)</i>		
Normalized planned time	Normalized planned time for project i at t	$\tau_{i,t} = \frac{t-s_i+1}{D_i^{\text{plan}}}$
Truncated normalized time	Normalized time restricted to $[0, 1]$	$\tau_{i,t}^c = \min\{1, \max\{0, \tau_{i,t}\}\}$
Beta shape parameters	Shape parameters of the Beta S-curve	$a_i, b_i > 0$
Cumulative expenditure fraction	Planned cumulative expenditure fraction	$F_i(\tau) = I_\tau(a_i, b_i)$
Planned cumulative expenditure	Planned cumulative expenditure at time t	$\bar{E}_{i,t} = \text{BAC}_i F_i(\tau_{i,t}^c)$
Planned period expenditure	Planned expenditure during period t	$\bar{e}_{i,t} = \bar{E}_{i,t} - \bar{E}_{i,t-1}$
<i>Budget and Credit</i>		
Available budget	Available internal portfolio budget at t	B_t^{avail}
Credit used	Credit used at time t	c_t
Outstanding debt	Outstanding debt at start of period t	D_t
Credit interest rate	Credit interest rate at t	r_t
Debt repayment	Debt repayment at t	q_t
<i>Mandatory Allocation</i>		
Mandatory allocation	Minimum mandatory allocation for project i at t	$m_{i,t}$
Baseline mandatory allocation	Mandatory allocation before intervention	$m_{i,t}^0$
<i>Project Progress and State</i>		
Planned progress	Planned cumulative progress for project i at t	$P_{i,t}^{\text{plan}}$
Actual progress	Actual cumulative progress for project i at t	$P_{i,t}^{\text{actual}}$
Performance gap	Difference between planned and actual progress	$\Delta_{i,t} = P_{i,t}^{\text{plan}} - P_{i,t}^{\text{actual}}$
Actual cumulative expenditure	Actual cumulative expenditure for project i at t	$E_{i,t}$
Schedule performance index	Schedule performance index	$\text{SPI}_{i,t}$
Cost performance index	Cost performance index	$\text{CPI}_{i,t}$
Remaining duration	Remaining duration estimate	$RD_{i,t}$
<i>Intervention Mechanism</i>		
Intervention threshold	Intervention activation threshold	τ_i
Intervention indicator	Endogenous intervention indicator	$I_{i,t} \in \{0, 1\}$
Intervention effectiveness	Intervention effectiveness parameter	$\kappa_{i,t}$
Intervention allocation increase	Additional allocation induced by intervention	χ_i
<i>Performance Dynamics</i>		
Baseline efficiency	Budget-to-progress conversion efficiency	$\alpha_{i,t}$
Adjusted efficiency	Intervention-adjusted efficiency	$\tilde{\alpha}_{i,t}$
Performance disturbance	Stochastic disturbance	$\varepsilon_{i,t}$
<i>Revenue and Milestones</i>		
Milestone set	Set of milestones for project i	$\mathcal{M}_i$
Milestone threshold	Progress threshold for milestone j	$\theta_{i,j}$
Payment fraction	Payment fraction for milestone j	$\phi_{i,j}$
Payment amount	Payment amount for milestone j	$R_{i,j}$
Milestone revenue	Total milestone revenue at t	V_t
<i>Objective-Related</i>		
Period performance metric	Period performance metric	R_t
Discount factor	Discount factor	$\gamma \in (0, 1]$
Delay penalty weight	Weight assigned to delay penalties	α_D
Credit financing cost	Credit financing cost	C_t^{credit}
Delay penalty cost	Liquidated damages and delay penalties	C_t^{delay}
Other portfolio costs	Other portfolio costs	C_t^{other}

4 Methodology

4.1 Overview of the Proposed Solution Framework

4.2 Literature-Calibrated Synthetic Portfolio Generation

4.3 Environment Initialization

4.4 MDP Specification

4.5 State Representation

4.6 Action Space and Feasibility Handling

4.7 Transition Dynamics

4.8 Reward Function

4.9 PPO Training Procedure

4.10 Benchmark Policies and Evaluation Metrics

4.11 Forward Policy Rollout

5 Results and Analysis

5.1 Overview

[This section will be completed in future work after experimental implementation. Placeholder structure provided below.]

5.2 Solution Quality Comparison

[To be completed: Tables and figures showing:]

- Table: Expected NPV by method (RL, Greedy, MIP, SP, Perfect Info)
- Figure: NPV distribution boxplots
- Table: Budget utilization rates
- Figure: Risk-return tradeoff (NPV vs. variance)

Expected findings:

- RL outperforms deterministic MIP by $\sim 15\text{--}20\%$ in expected NPV
- RL matches or exceeds stochastic programming performance
- RL achieves $\sim 90\text{--}95\%$ of perfect-information upper bound

5.3 Computational Performance

[To be completed: Tables and figures showing:]

- Table: Solve time by portfolio size and method
- Figure: Scalability plot (log-log scale)

- Table: Inference latency statistics (mean, P95, P99)

Expected findings:

- RL inference: 100–200ms per decision (constant with N)
- MIP solve time: 15–45 minutes for $N = 10$, exponential growth
- Speedup factor: $10,000\times$ for $N = 10$, increasing with N

5.4 Sensitivity Analysis

[To be completed: Analysis of performance under varying conditions:]

Uncertainty Level:

- RL maintains robust performance across low/medium/high uncertainty
- Deterministic MIP degrades significantly under high uncertainty
- SP performance depends on scenario tree quality

Budget Constraint Tightness:

- RL learns effective prioritization under tight budgets
- Greedy heuristic fails to optimize across projects
- MIP optimal under loose budgets but slow to compute

Portfolio Composition:

- Performance consistent across 60/40 domestic/international mix
- Sensitivity to extreme compositions (80/20, 20/80) to be analyzed

5.5 Ablation Studies

[To be completed: Component-wise analysis:]

- Impact of rolling horizon length $H \in \{6, 12, 18, 24\}$
- Effect of reward function components (NPV, utilization, penalties)
- Importance of SPI-CPI correlation in training data
- Value of attention mechanism (if implemented)

5.6 Discussion

[To be completed: Interpretation of results, including:]

Why RL Outperforms Classical Methods:

- Learns robust policies from stochastic simulations
- Adapts to partial observability without explicit transition probabilities
- Generalizes across portfolio configurations

Computational Advantage:

- One-time training cost amortized over many deployments
- Inference time independent of portfolio size (within trained range)
- Enables real-time scenario analysis and Monte Carlo simulation

Limitations:

- Performance bounded by training data distribution
- Requires fine-tuning for company-specific patterns
- Black-box nature may limit practitioner trust (addressed by interpretability analysis)

6 Conclusion

6.1 Summary of Contributions

This paper establishes the foundational methodology for Reinforcement Learning-based project portfolio budgeting under cashflow uncertainty. Our work addresses a critical gap in operations research: the inability of classical optimization methods to handle sequential decision-making under partial observability in portfolio management.

We make four primary contributions:

(1) Literature-Calibrated Synthetic Data Generation: We developed a comprehensive framework for generating realistic project portfolios with empirically-grounded uncertainty distributions. All parameters are calibrated from published studies (318 megaprojects, multiple empirical analyses), ensuring scientific validity while addressing confidentiality constraints.

(2) Rolling Horizon Control Architecture: We proposed a rolling horizon framework that maintains fixed 12-period episodes while handling variable-duration portfolios through sequential re-planning, preventing state-space explosion and enabling tractable RL training.

(3) Formal Problem Formulation: We provided the first rigorous mathematical formulation of portfolio budgeting as a POMDP with explicit state space, action space, transition dynamics, and reward function, all grounded in empirical project management literature.

(4) Complete Modeling Framework: We documented a modular portfolio modeling system covering composition, cashflows, uncertainty, and revenue plans, with all assumptions explicitly stated and future extensions identified.

6.2 Implications for Research and Practice

For Operations Research: This work demonstrates that RL can address problems where classical OR methods fail due to unknown transition probabilities and computational intractability. The rolling horizon architecture provides a template for applying RL to other long-horizon sequential decision problems.

For Project Management: The literature-calibrated synthetic data methodology enables reproducible research in domains with confidentiality constraints. Our framework can be extended to construction, infrastructure, R&D, and other project-intensive industries.

For Machine Learning: This application showcases RL’s potential in operations research domains, complementing recent successes in robotics, game playing, and control. The integration of domain knowledge (S-curves, milestone payments, SPI/CPI dynamics) with data-driven learning exemplifies hybrid AI approaches.

6.3 Limitations and Scope Boundaries

We deliberately adopted the *Simplest Model Principle* to establish a foundational framework. Key limitations include:

- **Aggregated cashflow modeling:** No resource-level detail (labor, materials, equipment)
- **Fixed portfolio composition:** 60/40 domestic/international mix assumed optimal
- **Milestone-based payments only:** Alternative structures (time-based, cost-plus) excluded
- **No project interdependencies:** Shared resources and technology spillovers not modeled
- **Synthetic data only:** Real-world validation pending company partnerships

Each limitation represents a natural extension for future work, documented with clear research directions.

6.4 Future Work

Immediate Next Steps (Current Research):

1. **RL Agent Implementation:** Select and train policy network (PPO, SAC, or Transformer-based)
2. **Baseline Comparison:** Implement MIP, stochastic programming, and greedy heuristics
3. **Experimental Evaluation:** Conduct comprehensive experiments across scenarios
4. **Sensitivity Analysis:** Analyze robustness to parameter variations

Medium-Term Extensions (6–12 Months):

1. **Fine-Tuning Protocol:** Develop methodology for adapting pretrained models to company-specific data
2. **Interpretability Analysis:** Apply attention mechanisms or SHAP values to explain allocation decisions
3. **Multi-Objective Optimization:** Extend to Pareto-optimal policies balancing NPV, risk, and strategic alignment
4. **Real-World Validation:** Partner with EPC contractors for pilot deployment

Long-Term Research Directions (1–3 Years):

1. **Resource-Level Modeling:** Disaggregate cashflows into labor, materials, equipment with detailed scheduling
2. **Dynamic Portfolio Selection:** Integrate project selection with execution optimization
3. **Multi-Agent RL:** Model interactions between contractor, client, and subcontractors
4. **Transfer Learning:** Investigate cross-domain transfer (oil & gas $\rightarrow$ construction $\rightarrow$ infrastructure)
5. **Hierarchical RL:** Develop two-level policies (strategic portfolio allocation + tactical project control)

6.5 Broader Impact

This research has potential impact beyond project portfolio management:

Methodological Contributions:

- Template for applying RL to OR problems with unknown transition dynamics
- Framework for literature-calibrated synthetic data generation
- Rolling horizon architecture for long-horizon sequential decision problems

Practical Applications:

- Real-time decision support for portfolio managers
- Scenario analysis for strategic planning
- Risk assessment through Monte Carlo simulation
- Training tool for project management education

Open Science:

- Complete codebase and datasets released as open source
- Reproducible research enabling community validation and extension
- Benchmark problem for RL algorithm development

6.6 Concluding Remarks

Project portfolio budgeting under uncertainty represents a critical challenge for capital-intensive industries. Traditional optimization methods fail due to computational intractability and unknown state transition probabilities. This paper establishes the foundational methodology for addressing this challenge through Reinforcement Learning with rolling horizon control.

By developing a literature-calibrated synthetic data generation framework, we overcome the confidentiality barrier that has prevented progress in this domain. Our formal problem formulation and complete modeling framework provide a rigorous foundation for future research.

The path forward is clear: implement and evaluate RL agents, compare against classical baselines, and develop fine-tuning protocols for company-specific deployment. We invite the research community to build upon this foundation, extending the methodology to related domains and advancing the state of the art in RL-based operations research.

The complete modeling framework, instance generator, and documentation will be released as open-source software upon publication, enabling reproducible research and practical deployment in this emerging field.

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A Mathematical Proofs

[Placeholder for mathematical proofs. To be completed if needed.]

A.1 Proof of Rolling Horizon Optimality Conditions

[To be added: Conditions under which rolling horizon policy approximates infinite-horizon optimal policy]

A.2 Proof of State Space Completeness

[To be added: Demonstration that state representation captures all decision-relevant information]

B Parameter Calibration Details

B.1 Complete Parameter Tables

This appendix provides comprehensive parameter tables for all model components, with literature citations and calibration details.

Table 2: Portfolio Composition Model Parameters

Parameter	Value	Unit	Source
Domestic weight	0.60	fraction	Khanzadi et al. (2018)
International weight	0.40	fraction	Khanzadi et al. (2018)
Portfolio size min	8	projects	Merrow (2011), CII (2019)
Portfolio size max	18	projects	Merrow (2011), CII (2019)

Table 3: Profit Margin Distribution Parameters

Parameter	Domestic	International	Source
Mean μ	0.10	0.14	Khanzadi et al. (2018)
Std Dev σ	0.012	0.030	Khanzadi et al. (2018)
Lower bound a	0.08	0.12	Iran Budget Law
Upper bound b	0.12	0.16	Market analysis

B.1.1 Portfolio Composition Parameters**B.1.2 Profit Margin Parameters****B.1.3 Project BAC Parameters****B.1.4 S-Curve Parameters****B.1.5 Performance Uncertainty Parameters****B.1.6 Duration Model Parameters****B.1.7 Milestone Payment Structure****B.2 Validation Results**

[To be completed: Statistical validation of generated instances against target distributions]

B.2.1 Distributional Validation

[Tables showing sample statistics vs. target values for all key variables]

B.2.2 Correlation Validation

[Correlation matrices comparing generated data to empirical values]

B.2.3 Portfolio-Level Validation

[Aggregate statistics for generated portfolios]

C Instance Generator Algorithm**C.1 Instance Generator Algorithm**

This appendix provides the complete pseudocode for the portfolio instance generator.

Table 4: Project BAC Lognormal Distribution

Parameter	Value	Source
$\mu_{\log}$	5.01	Merrow (2011)
$\sigma_{\log}$	0.69	Merrow (2011)
Median BAC	\$150M	Merrow (2011)
Mean BAC	\$280M	Merrow (2011)
Min BAC	\$50M	Industry practice
Max BAC	\$5B	Industry practice

Table 5: S-Curve Beta Distribution Base Parameters

Parameter	Value	Source
α_{base}	2.5	Cioffi (2005)
β_{base}	2.0	Cioffi (2005)
Peak time τ^*	0.60	Kenley & Wilson (1986)
Size exponent	0.15	Barraza et al. (2007)
Duration exponent	0.12	Barraza et al. (2007)
Residual noise σ	0.10	Calibrated

Algorithm 1 Portfolio Instance Generator**Require:** Portfolio size range $[N_{\min}, N_{\max}]$, parameter tables**Ensure:** Portfolio instance $\mathcal{P}$ with all attributes

```

1:
2: Stage 1: Portfolio Composition
3:  $N \leftarrow \text{UniformInt}(N_{\min}, N_{\max})$ 
4:  $N_{\text{domestic}} \leftarrow \lfloor 0.6 \times N \rfloor$ 
5:  $N_{\text{international}} \leftarrow N - N_{\text{domestic}}$ 
6:
7: Stage 2: Project Characteristics
8: for  $i = 1$  to  $N$  do
9:   Assign category: domestic or international
10:  Sample  $\log(\text{BAC}_i) \sim \mathcal{N}(\mu_{\log}, \sigma_{\log})$ 
11:  Clip  $\text{BAC}_i$  to  $[\$50M, \$5B]$ 
12:  Sample  $\pi_i \sim \text{TruncatedNormal}(\mu_{\pi}, \sigma_{\pi}, a_{\pi}, b_{\pi})$ 
13:  Compute  $R_i \leftarrow \text{BAC}_i \times (1 + \pi_i)$ 
14:  Sample  $D_i \leftarrow D_{\text{base}} \times (\text{BAC}_i / \text{BAC}_{\text{ref}})^{0.35} \times \epsilon_D$ 
15:  Clip  $D_i$  to  $[12, 60]$  months
16: end for
17:
18: Stage 3: S-Curve Parameters
19: for  $i = 1$  to  $N$  do
20:   Compute size multiplier:  $m_{\text{size}} \leftarrow (\text{BAC}_i / \text{BAC}_{\text{median}})^{0.15}$ 
21:   Compute duration multiplier:  $m_{\text{dur}} \leftarrow (D_i / D_{\text{median}})^{0.12}$ 
22:   Get category multiplier  $m_{\text{cat}}$  (0.9 for domestic, 1.1 for international)
23:   Sample risk level: low or high
24:   Get risk multipliers  $m_{\text{risk}}^{\alpha}, m_{\text{risk}}^{\beta}$ 
25:   Sample  $\epsilon_{\alpha}, \epsilon_{\beta} \sim \text{LogNormal}(0, 0.10)$ 
26:    $\alpha_i \leftarrow \alpha_{\text{base}} \times m_{\text{size}} \times m_{\text{cat}} \times m_{\text{risk}}^{\alpha} \times \epsilon_{\alpha}$ 
27:    $\beta_i \leftarrow \beta_{\text{base}} \times m_{\text{dur}} \times m_{\text{cat}} \times m_{\text{risk}}^{\beta} \times \epsilon_{\beta}$ 
28:   Clip  $\alpha_i$  to  $[1.2, 5.0]$ ,  $\beta_i$  to  $[1.2, 5.5]$ 
29: end for
30:
31: Stage 4: Cashflow Profiles
32: for  $i = 1$  to  $N$  do
33:   for  $t = 0$  to  $D_i$  do

```

Table 6: SPI and CPI Distribution Parameters

Parameter	SPI	CPI	Correlation	Source
α	2.8	3.2	–	Merrow (2011)
β	2.2	1.8	–	Love et al. (2016)
Lower bound	0.3	0.5	–	Empirical range
Upper bound	1.3	1.2	–	Empirical range
Mean	0.86	0.87	–	Calibrated
Std Dev	0.18	0.14	–	Calibrated
$\rho(\text{SPI, CPI})$	–	–	0.68	Love et al. (2016)

Table 7: Project Duration Model Parameters

Parameter	Value	Source
D_{base}	24 months	Merrow (2011)
BAC_{ref}	\$150M	Merrow (2011)
Scale exponent	0.35	Merrow (2011)
Noise σ_D	0.15	Calibrated
Min duration	12 months	Industry practice
Max duration	60 months	Industry practice

Table 8: Standard Milestone Payment Weights

Milestone	Progress Threshold	Payment Weight
Advance	0%	10%
Milestone 1	25%	20%
Milestone 2	50%	30%
Milestone 3	75%	30%
Final	100%	10%

C.2 Validation Algorithm

Algorithm 2 Instance Validation

Require: Generated portfolio instance $\mathcal{P}$, target statistics

Ensure: Validation report (pass/fail for each check)

```

1:
2: Level 1: Distributional Validation
3: for each variable  $X$  in  $\{\text{BAC}, \pi, D, \alpha, \beta, \text{SPI}, \text{CPI}\}$  do
4:   Compute sample mean  $\bar{X}$ , std dev  $s_X$ 
5:   Check:  $|\bar{X} - \mu_{\text{target}}|/\mu_{\text{target}} < 0.05$  ▷ 5% tolerance
6:   Check:  $|s_X - \sigma_{\text{target}}|/\sigma_{\text{target}} < 0.10$  ▷ 10% tolerance
7: end for
8:
9: Level 2: Correlation Validation
10: Compute  $\rho(\text{SPI}, \text{CPI})$ 
11: Check:  $|\rho - 0.68| < 0.05$ 
12:
13: Level 3: Portfolio-Level Validation
14: Compute total portfolio BAC:  $\sum_{i=1}^N \text{BAC}_i$ 
15: Compute domestic fraction:  $\sum_{i \in \text{domestic}} \text{BAC}_i / \sum_{i=1}^N \text{BAC}_i$ 
16: Check: domestic fraction  $\in [0.55, 0.65]$  ▷ 60%  $\pm$  5%
17:
18: Level 4: Sanity Checks
19: for  $i = 1$  to  $N$  do
20:   Check:  $\text{BAC}_i > 0, D_i > 0, \pi_i > 0$ 
21:   Check:  $\alpha_i \in [1.2, 5.0], \beta_i \in [1.2, 5.5]$ 
22:   Check:  $\text{SPI}_i \in [0.3, 1.3], \text{CPI}_i \in [0.5, 1.2]$ 
23:   Check:  $\sum_m w_m = 1.0$  ▷ Milestone weights sum to 100%
24: end for
25:
26: return Validation report

```

C.3 Implementation Notes

Computational Complexity: The instance generator has time complexity $O(N \times D_{\max})$ where N is portfolio size and $D_{\max}$ is maximum project duration. For typical values ($N = 15, D_{\max} = 60$), generation takes < 1 second per instance.

Reproducibility: All random number generation uses seeded PRNGs (NumPy default_rng with explicit seeds). Seeds are logged for each generated instance.

Parallelization: Instance generation is embarrassingly parallel. For large datasets (10,000+ instances), use multiprocessing with one worker per CPU core.

Output Format: Instances are serialized as JSON or HDF5 files with standardized schema for interoperability.